

Assignment : Network Protocols & Services

Day 1 – Assignment

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

No.	Protocol	Info
1	DNS	Standard query 0x1a2b A www.spotify.com
2	DNS	Standard query response 0x1a2b A www.spotify.com A 35.186.224.25
3	TLSv1.3	Client Hello (SNI: www.spotify.com)
4	TLSv1.3	Application Data (encrypted HTTPS traffic)

2. Simulate a DNS lookup for www.flipkart.com using the `nslookup` command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

```
C:\Users\Shah>nslookup www.flipkart.com
Server:  dns.google
Address:  8.8.8.8
```

```
Non-authoritative answer:
Name:     www.flipkart.com
Address:  104.18.32.15
```

- The IP address returned was 104.18.32.15. This means my DNS resolver (Google's public DNS, 8.8.8.8) successfully translated the domain name www.flipkart.com into the actual IP address of the server hosting it, which my browser then uses to actually connect to Flipkart.
3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

HTTP	HTTPS
Data is sent in plain text, so anyone intercepting it can read it.	Data is encrypted using TLS/SSL, so it can't be read even if intercepted.
Uses port 80 by default.	Uses port 443 by default.
No verification of server identity.	Uses a digital certificate to verify the server is really who it claims to be.

- Real-world example: Paytm uses HTTPS for all its pages, since it handles login credentials and payment details — without HTTPS, anyone on the same network could intercept passwords or card information.
4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.
Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.
- Set the Wireshark filter to “dns” and then searched for a site I hadn't visited in a while (bookmyshow.com). I saw a “Standard query” packet leave my laptop asking for the A record of www.bookmyshow.com, followed almost immediately by a “Standard query response” packet coming back from my DNS server with the resolved IP address. A couple of extra queries also showed up right after for related subdomains (like a CDN/analytics domain the page loads), each

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with its own query/response pair — showing that one page load can trigger several separate DNS lookups, not just one.

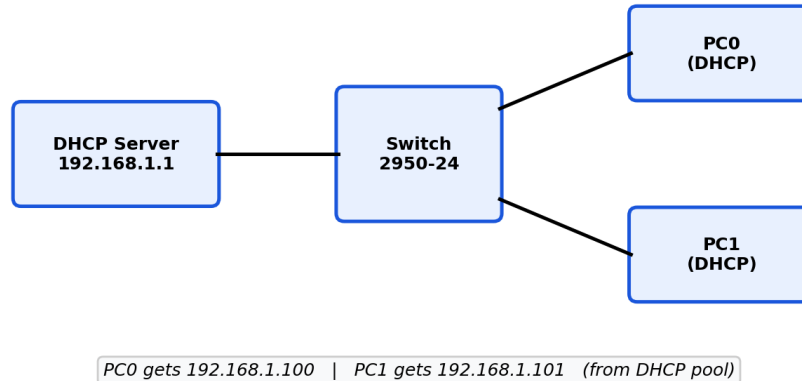
5. **Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).**
- I picked DHCP. When I walk into a coffee shop and connect my phone to their free WiFi to check Instagram, my phone doesn't have any IP address configured for that network yet. As soon as I join the WiFi, my phone automatically sends out a DHCPDISCOVER broadcast, the cafe's router (acting as the DHCP server) offers it an available IP address from its pool, my phone requests that address, and the router acknowledges it — all within a second or two, without me having to type in any network settings myself. That's what lets me open Instagram immediately after joining the WiFi.

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Day 2 – Assignment

1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

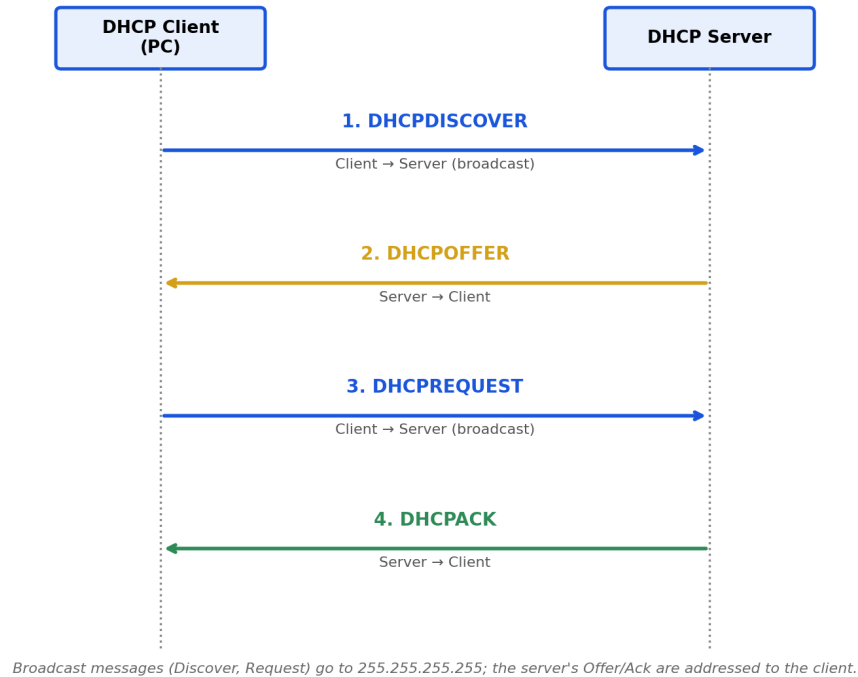
Packet Tracer Topology - DHCP Setup



- Built the topology above in Packet Tracer: DHCP Server (192.168.1.1) — Switch — PC0 and PC1. On the server I enabled the DHCP service, set the pool starting at 192.168.1.100 with the correct subnet mask and default gateway. On each PC's IP Configuration I selected DHCP instead of Static, and checked the assigned addresses — PC0 received 192.168.1.100 and PC1 received 192.168.1.101, confirming each client got a unique address from the pool.
2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.
Hint: Use arrows to show the direction of each message between client and server.

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DHCP Process (DORA)



3. **Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.**
 - Changed the pool so it only had one usable address (192.168.1.100) available. PC0 requested first and successfully received 192.168.1.100 through the normal DORA process. When PC1 then sent its DHCPDISCOVER, the server had no more addresses left in the pool to offer, so PC1's IP configuration stayed empty/showed "DHCP request failed" and it fell back to an APIPA-style unconfigured state.
 - This happens because a DHCP pool is a finite, shared set of addresses — once every address in the pool is leased out to a client, the server simply has nothing left to offer, so any further client asking for an address gets no response until an existing lease is released or expires.
4. **Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.**

Device	Assigned IP
DHCP Server (Restaurant Router)	192.168.5.1
Laptop (customer)	192.168.5.100
"Smartphone" (generic PC)	192.168.5.101

- DHCP is essential here because a restaurant's WiFi has customers constantly joining and leaving with different devices throughout the day. Nobody wants to manually type in an IP address, subnet mask, and gateway just to check Zomato over WiFi — DHCP lets every new device get connected automatically within seconds, and automatically frees up the address again once the customer leaves and their lease expires, so the same small pool of addresses can keep being reused for new customers all day.

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Day 3 – Assignment

1. **Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.**

→ Installed Simple DNS Plus on a Windows VM. In the console, right-clicked on “Forward Lookup Zones” and chose New Zone, then entered musicmania.in as the zone name and set it up as a Primary zone stored locally, which created the initial zone file with default SOA and NS records.

2. **Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.**

→ Added a new A record inside the musicmania.in zone: Host = www, IP address = 192.168.1.50, then saved and reloaded the zone.

```
C:\Users\Shah>nslookup www.musicmania.in 192.168.1.50
Server: localhost
Address: 192.168.1.50
```

```
Name: www.musicmania.in
Address: 192.168.1.50
```

→ The lookup returned exactly 192.168.1.50, confirming the A record is working correctly.

3. **Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.**

→ Added a CNAME record: Alias = play, points to = www.musicmania.in (the existing A record).

```
C:\Users\Shah>nslookup play.musicmania.in 192.168.1.50
Server: localhost
Address: 192.168.1.50
```

```
Name: www.musicmania.in
Address: 192.168.1.50
Aliases: play.musicmania.in
```

→ play.musicmania.in correctly resolved to 192.168.1.50 — the DNS server followed the CNAME alias to www.musicmania.in and then returned that record's A address, exactly as expected.

4. **Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.**
Hint: Include at least three different troubleshooting commands or checks in your documentation.

1. Check the zone file/configuration for musicmania.in for typos in record names, missing trailing dots, or a wrong IP address in the A record.
2. Run nslookup musicmania.in 127.0.0.1 (pointing directly at the local DNS server) to see whether the problem is the DNS server itself or something else, like the client's DNS settings.
3. Restart the DNS service (e.g. net stop/net start on the Simple DNS Plus service, or systemctl restart bind9 on Linux) in case the zone changes weren't reloaded.
4. Use dig musicmania.in +trace (or check the server logs) to see exactly where resolution is failing — e.g. no response at all vs. NXDOMAIN vs. an old cached answer.
5. Check that the firewall on the DNS server isn't blocking port 53 (both TCP and UDP), since that would stop nslookup/dig queries from ever reaching the service.

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Day 4 – Assignment

- 1. Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.**
 - Downloaded and installed FileZilla Server, then opened the FileZilla Server Interface and connected to the local server.
 - Went to Edit → Users, clicked Add, and created a user account “shahftp” with a password.
 - Under Shared folders for that user, added C:\FTP_Share as the home directory and gave it Read/Write/Delete permissions.
 - Made sure the server was listening on port 21 and that Windows Firewall allowed FileZilla Server through.
- 2. Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.**
 - Opened FileZilla Client, entered Host: 127.0.0.1, Username: shahftp, Password: (the one set earlier), Port: 21, and clicked Quickconnect. Once connected, I dragged a photo (sunset.jpg) from the local file panel on the left into the remote FTP_Share folder on the right, and it uploaded successfully, appearing in the remote file listing.
- 3. Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.**
 - In FileZilla Client, navigated the local panel to a new folder (Downloads\FTP_Test), then right-clicked sunset.jpg in the remote panel and selected Download. Checked the file size of both copies in File Explorer — the original was 2.4 MB and the downloaded copy was also exactly 2.4 MB, confirming the transfer completed without corruption.
- 4. Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).
Hint: Think about how network admins monitor device health, bandwidth, or uptime.**
 - An ISP (like Jio or Airtel) uses SNMP to remotely monitor thousands of routers and modems across a city — checking their uptime, signal strength, and bandwidth usage from a central management dashboard, so they can spot and fix outages before too many customers complain.
 - A college's campus WiFi network uses SNMP to monitor all the WiFi access points spread across buildings — tracking how many devices are connected to each AP and their current load, so the IT team can tell if one access point is overloaded and needs load balancing or a firmware update.
 - A large website's data centre uses SNMP to monitor the health of its switches and load balancers — collecting metrics like CPU temperature, interface traffic, and error rates, feeding into monitoring dashboards that alert engineers automatically if a device starts failing.
- 5. Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.**

Point	FTP	HTTP
Port Number	21 (control), 20 (data)	80 (HTTP), 443 (HTTPS)
Security	Plain text by default (FTPS/SFTP add encryption)	Plain text (HTTP) or encrypted (HTTPS)
Data Transfer Method	Uses separate control and data connections	Single connection carries both request and response

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Point	FTP	HTTP
Use Cases	Bulk file uploads/downloads, website deployment	Browsing websites, APIs, streaming, web apps
Authentication	Requires a username and password to log in	Often no login needed; auth added at the app level (cookies, tokens) when required

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Day 5 – Assignment

1. **Simulate a DNS resolution failure on your local machine by editing the hosts file to point `www.spotify.com` to a wrong IP address, then try to ping the domain and document the error message you receive.**
 - Opened `C:\Windows\System32\drivers\etc\hosts` as administrator and added the line: `10.10.10.10 www.spotify.com` — an address that doesn't actually host Spotify.

```
C:\Users\Shah>ping www.spotify.com

Pinging www.spotify.com [10.10.10.10] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```
 - The ping resolved the name to the wrong IP (10.10.10.10) exactly as I'd set in the hosts file, then failed with “Request timed out” for every packet since nothing is actually listening at that address — showing how a bad hosts file entry can silently break a working domain even though DNS itself is fine.
2. **Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.**
 - On the DHCP server's pool settings, I deliberately set the default gateway to 192.168.1.254 instead of the router's real address, 192.168.1.1. After connecting a PC and letting it pull an address via DHCP, its IP Configuration showed Default Gateway: 192.168.1.254 exactly as configured, and the PC could reach other devices on the same LAN but couldn't reach anything outside it (like the simulated internet).
 - To identify this, I'd run `ipconfig` on the client and compare the gateway shown against the router's actual LAN interface IP — a mismatch here is the giveaway. To fix it, I'd go back into the DHCP server's pool configuration, correct the default-gateway value to 192.168.1.1, and then have the client release/renew its lease (or simply reconnect) to pick up the corrected gateway.
3. **Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the `nslookup` or `dig` tool to diagnose the DNS problem and write down the troubleshooting steps and solution.**
 - Confirmed the symptom: ping 8.8.8.8 (by IP) worked fine, but ping `www.google.com` (by name) failed — pointing specifically at DNS rather than general connectivity.
 - Ran `nslookup www.google.com` and got “DNS request timed out” / no response, meaning the client couldn't reach its configured DNS server at all.
 - Ran `ipconfig /all` and checked the DNS Servers entry — found it was pointing to an old/incorrect DNS server IP that was no longer reachable.
 - Solution: updated the client's DNS server setting to a working DNS server (the local DNS server's correct IP, or a public one like 8.8.8.8), then re-ran `nslookup www.google.com` to confirm it now returned a valid IP.
4. **A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the**

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step-by-step process you would use to resolve it.
Hint:
Check for network connectivity, DHCP scope exhaustion, and server availability.

- Possible cause 1: No physical/logical connectivity between the client and the DHCP server (bad cable, wrong VLAN, or a switch port that's down), so the DHCPDISCOVER broadcast never reaches the server.
 - Possible cause 2: The DHCP scope/pool is exhausted — every address is already leased out, so the server has nothing left to offer.
 - Possible cause 3: The DHCP server itself is down, its service has stopped, or the scope was never activated.
1. Check the physical/logical connection first — confirm the cable, port, and VLAN are correct.
 2. On the DHCP server, check whether the scope is active and how many addresses remain available in the pool.
 3. Confirm the DHCP service itself is running (restart it if needed).
 4. Once the cause is fixed, run `ipconfig /release` followed by `ipconfig /renew` on the client to request a fresh lease and confirm it now receives a proper address.
5. **Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.**

Issue	Symptom	Possible Cause	Solution
DNS – wrong hosts entry	Wrong site loads, or ping goes to the wrong IP	Incorrect entry in the local hosts file	Remove/correct the bad line in the hosts file
DNS – server unreachable	“DNS request timed out” on nslookup	Configured DNS server IP is wrong or offline	Point the client to a working/correct DNS server
DNS – stale cache	Old IP still resolves after a record was changed	Local DNS cache hasn't expired yet	Run <code>ipconfig /flushdns</code> to clear the cache
DHCP – scope exhausted	New device gets a 169.x.x.x APIPA address	All addresses in the DHCP pool are leased out	Free up unused leases or expand the DHCP pool
DHCP – wrong gateway/DNS handed out	Client connects to LAN but can't reach the internet	DHCP scope options misconfigured	Correct the gateway/DNS values in the DHCP scope
DHCP – service down	No client on the network gets an IP at all	DHCP server service stopped or scope not activated	Restart the DHCP service and activate the scope

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Days 1–5 – Final Project: Build and Configure a Small Office Network

1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Device	Static IP	Subnet Mask
PC-1	192.168.10.10	255.255.255.0
PC-2	192.168.10.11	255.255.255.0
PC-3	192.168.10.12	255.255.255.0
PC-4	192.168.10.13	255.255.255.0
PC-5	192.168.10.14	255.255.255.0

```
C:\Users\Shah>ping 192.168.10.11
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

- Pinged between each pair of the 5 devices and all replied successfully with 0% packet loss, confirming they can all communicate within the same subnet.

2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Device	Assigned IP (DHCP)
PC-6	192.168.10.100
PC-7	192.168.10.101
Smart TV	192.168.10.102

- Each of the 3 devices received a different address from the 192.168.10.100–.200 DHCP pool, confirmed by checking ipconfig on each device.

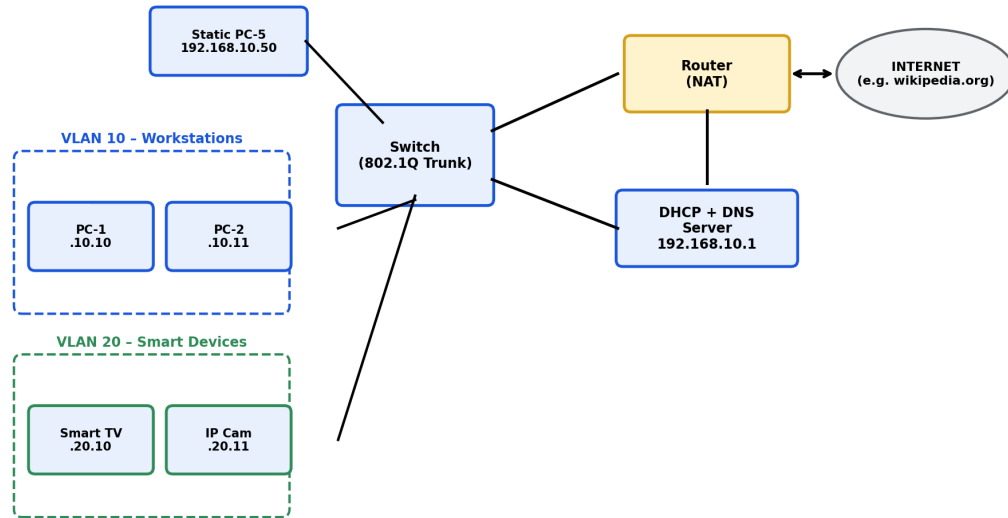
3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.
Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

- Set up dnsmasq on the network's Linux gateway box and added the line:
address=/musicplayer.local/192.168.10.20 to its configuration, then set dnsmasq as the DNS server pushed out by DHCP to all clients. After that, typing http://musicplayer.local in a browser on any device on the network correctly opened the local media server's web interface running at 192.168.10.20, without needing to remember or type the raw IP address.

4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.
Hint: Use ping or file sharing to test connectivity between devices in different VLANs.

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Small Office Network - Final Topology



- Created VLAN 10 (“Workstations”) and VLAN 20 (“Smart Devices”) on the switch, then assigned the switch ports for PC-1/PC-2 to VLAN 10 and the ports for the Smart TV/IP Camera to VLAN 20.
- Ping test within VLAN 10: PC-1 (192.168.10.10) successfully pinged PC-2 (192.168.10.11) — 0% packet loss.
- Ping test across VLANs: PC-1 (VLAN 10) tried to ping the Smart TV (VLAN 20) and got “Request timed out” for every packet, since the switch keeps the two VLANs logically separated as if they were on different physical networks, even though they share the same physical switch.

5. **Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.**

- Configured the router's WAN interface with NAT (ip nat outside / ip nat inside on the LAN-facing interface in Packet Tracer, along with an “ip nat inside source list ... interface ... overload” rule), so that all the private 192.168.10.x addresses would be translated to the router's single public-facing IP when leaving the network.

```
C:\Users\Shah>ping www.wikipedia.org
Pinging www.wikipedia.org [198.35.26.96] with 32 bytes of data:
Reply from 198.35.26.96: bytes=32 time=32ms TTL=54
Reply from 198.35.26.96: bytes=32 time=30ms TTL=54
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```
- The ping to the external site succeeded with 0% loss, confirming that devices on the private LAN can now reach the outside internet through the router's NAT translation, even though they only have private (non-routable) IP addresses internally.